

MS review:

Constrained evolution of a core winter proteome across independently

Independent evolution of freezing tolerance across five PACMAD grass lineages is governed by evolutionary constraint on protein-level response magnitude, with shared cold-responsive proteins showing higher cross-species fold-change correlation ($\rho = 0.80$) than background proteins ($\rho = 0.45$), while structural divergence of the universally conserved LEA3 protein in maize—specifically a variant third repeat that disrupts amphipathic balance—compromises cryoprotective function despite transcriptional induction to levels comparable with cold-tolerant relatives.

1. GAP 1 (MAJOR) No Matched LT50 Phenotypes for Sampled Clones

1.1. Background information

Cross-species correlations of winter/summer rhizome proteome fold-changes (Figure 2) and the universality of LEA3 elevation (Figure 3) are interpreted as evidence of evolutionary constraint linked to freezing tolerance. Sampling and TMT workflows (Figure 1, Table 1) provide robust seasonal datasets, and PCA shows clear seasonal separation. However, the study does not report matched quantitative tolerance metrics (e.g., LT50, survival assays) for those same clones and tissues.

1.2. Identified gap

The study does not provide matched quantitative freezing tolerance phenotypes (e.g., LT50, survival) for the sampled clones/tissues, so constrained proteome responses cannot be causally linked to freezing tolerance across species. This

methodological gap is widespread in the field - comparative proteomic studies typically use genotypes with known tolerance differences [1][2] rather than quantifying tolerance in the actual samples analyzed. Without LT50 or survival assays in the same plants/tissues, cross-species fold-change correlations and LEA3 universality remain associations with winter dormancy rather than demonstrated drivers of tolerance. This affects the primary evolutionary interpretation and any translational recommendations.

1.3. Suggested strategies

1.3.1. Suggested textual edits

To address this gap through paper edits, we suggest the following approach: First, justify the current findings by clarifying the ecological relevance and candidate-mechanism status. For example, add: "Our common-garden design ensured identical winter conditions across all genotypes (Fig. 1b), and the sampled lineages are temperate and freezing-tolerant by native ranges (Fig. 1e). Accordingly, the conserved response magnitudes we report represent candidate mechanisms for freezing survival in rhizomes." Next, acknowledge the limitation: "We did not measure quantitative freezing tolerance (e.g., LT50 or survival under controlled freezing ramps) in the exact clones and rhizome tissues profiled here; therefore, we interpret constrained proteome responses as candidate mechanisms pending direct phenotypic validation." Finally, modify the main claim to reflect this nuanced perspective: Replace "Independent evolution of freezing tolerance in PACMAD grasses is governed by evolutionary constraints on protein-level response magnitude" with "Independent evolution of freezing tolerance in PACMAD grasses is consistent with

constrained rhizome protein-level response magnitude; establishing causal linkage to tolerance requires matched phenotyping across the sampled clones."

1.3.2. Potential next steps

1.3.2.1. Option 1

Apply the existing freezing survival assay paradigm (-8°C for 3 h) to the same clonal plants and rhizome tissues profiled by TMT, and score survival/regrowth. Integrate these phenotypes with LEA3 and shared DAP fold-changes from the current proteomes to test molecular-to-phenotype linkage.

1.3.2.2. Option 2

Generate LT50 curves under controlled freezing ramps for each clone/species and pair these measurements with pre/post-acclimation rhizome proteomes. This establishes whether constrained response magnitudes quantitatively predict tolerance across independently adapted lineages.

2. GAP 2 (MINOR) No Cross-Batch Calibration of TMT Fold-Changes

2.1. Background information

Seasonal rhizome proteomes were quantified across multiple TMT batches with species-specific searches (Figure 1d, Table 1). Fold-change correlations across species (Figure 2b) support the evolutionary constraint claim. However, without a common reference channel or external spike-in standards, cross-batch/cross-species fold-change magnitudes may not be directly comparable.

2.2. Identified gap

The study lacks cross-species quantitative calibration (e.g., spike-in standards or common reference channels) across TMT batches, making fold-change magnitudes that underpin correlation analyses potentially non-comparable

across species. Total peptide normalization and species-specific database differences can alter dynamic range, and ratio compression varies by batch; without calibration, observed correlation contrasts could reflect technical harmonization rather than biological constraint.

2.3. Suggested strategies

2.3.1. Suggested textual edits

To address this gap through paper edits, we suggest the following approach: First, justify the current findings by explaining the mitigation steps already taken. For example, add: "We co-batched winter and summer samples for each species and used RTS-SPS-MS3 to reduce ratio compression, thereby improving within-batch quantitation." Next, acknowledge the limitation: "We did not include a global reference channel or external spike-in standards across TMT batches; cross-species fold-change magnitudes are therefore compared after total peptide amount normalization and species-specific database searches." Finally, modify the main claim to reflect calibration plans: Replace "Shared cold-responsive proteins showed higher cross-species fold-change correlation ($\rho = 0.$ than background proteins ($\rho = 0.$ " with "Across TMT batches normalized by total peptide amount, shared cold-responsive proteins showed higher cross-species fold-change correlation ($\rho = 0.$ than background proteins ($\rho = 0.$; planned inclusion of common reference channels/spike-ins will confirm magnitude comparability across species."

2.3.2. Potential next steps

To address the absence of cross-species quantitative calibration across TMT batches, we propose a two-part experimental plan:

2.3.2.1. Option 1

Re-normalize existing datasets using housekeeping-anchored and shared-protein-anchored strategies, and re-analyze correlations to assess robustness of the observed contrast between shared DAPs and background.

2.3.2.2. Option 2

Implement a common TMT reference channel and/or external spike-ins across new batches and search against a pan-PACMAD proteome to harmonize identification/quantitation. This establishes calibrated fold-change magnitudes and confirms the evolutionary constraint inference on a standardized measurement scale.

3. Main Claim 1 Seasonal rhizome proteomes from five PACMAD species show distinct winter and summer protein accumulation patterns with species-specific numbers of differentially abundant proteins

3.1. GAP 1 (MINOR) DAP Counts Confounded by Unequal Power

3.1.1. Background information

The comparison of how many proteins change between winter and summer is central to concluding that species differ in the extent of seasonal remodeling. The study provides clear evidence of seasonal separation (principal component analysis in Fig. S2) and robust volcano plots and counts of differentially abundant proteins summarized in Table 1, with additional detail in Fig. S4. The authors also note that one species has fewer identifications due to a draft genome (Table S2), indicating variable proteome coverage across species.

3.1.2. Identified gap

Species-to-species comparisons of the number of

differentially abundant proteins are not adjusted for unequal biological replication, detection depth, and reference proteome completeness, potentially confounding the reported species-specific counts. Table 1 reports markedly different sample sizes across species (e.g., two replicates for one species versus three or more for others) and different total proteins identified, while Table S2 documents substantial differences in genome completeness that can affect peptide assignment. These factors can change statistical power and identification rates, which in turn influence the number of significant proteins in volcano plots (Fig. S4) and the apparent seasonal separation (Fig. S2). Systematic evaluations of proteomics normalization methods demonstrate that technical factors like detection depth and sample size significantly impact differential expression analysis outcomes [1][2], while studies show that uneven detection depth creates persistent bias in cross-species comparisons even at high coverage [3]. Without normalizing or modeling these imbalances, the conclusion that species differ in the number of seasonal changes may partly reflect technical rather than biological differences.

3.1.3. Suggested strategies

3.1.3.1. Suggested textual edits

First, justify the current findings by emphasizing existing safeguards. For example, add: "winter and summer samples from each species always included within the same batch." and "Given the inherent biological variability of a multi-species field study, we applied a stringent threshold for differential abundance." Next, acknowledge the

limitation: "S. nutans, with a draft genome containing 26% ambiguous bases, yielded fewer identifications (Table S2), and unequal replication can affect DAP counts across species." Finally, modify the claim to reflect this nuanced perspective: Replace "Differentially abundant proteins (DAPs) varied across species, with 58–174 proteins significantly increased and 56–218 decreased in abundance under freezing conditions (Table 1)." with "Within each species, winter-summer shifts were robust; cross-species DAP counts are descriptive and may reflect detection and power differences (Table 1; Table S2)."

3.1.3.2. Potential next steps

To address Species-to-species comparisons of the number of differentially abundant proteins are not adjusted for unequal biological replication, detection depth, and reference proteome completeness:

3.1.3.2.1. Option 1

Implement power-equalized reanalyses using existing TMT datasets in R. Specifically: (i) restrict to 1:1:1:1:1 orthogroups detected in both seasons across all species and report per-species DAP proportions (significant/quantifiable); (ii) perform replicate downsampling to $n = 2$ per season for species with >2 replicates, repeating 1,000 times and summarizing DAP proportions; (iii) perform identification-depth rarefaction by downsampling proteins to a common number across species, repeating 1,000 times and recomputing DAP

proportions; and (iv) estimate permutation-based expected significant counts by shuffling winter/summer labels within species to derive observed-to-expected ratios. This will provide power- and depth-aware estimates of seasonal remodeling that can be fairly contrasted across species.

3.1.3.2.2. Option 2

Increase replication for the underpowered species (*S. nutans*) to ≥ 3 biological replicates per season in a new TMTpro 18-plex with within-batch winter/summer pairing, and enhance identifications by applying the demonstrated DIAMOND-based peptide remapping strategy (as used for *M. giganteus*) to *S. nutans* against a closely related Andropogoneae proteome. Re-run the full pipeline (normalization, ANOVA/BH) and repeat the power-equalized analyses. This establishes cross-species DAP proportions under matched replication and improved identification, providing the definitive basis for cross-species comparisons.

3.2. GAP 2 (MINOR) Field Design Confounds Cold and Dormancy Effects

3.2.1. Background information

Field collections captured dormancy vs active growth states (Figure 1b) and showed seasonally distinct proteomes by PCA (Fig. S2) and volcano plots (Fig. S4), with per-species DAP summaries in Table 1. The study infers that contrasts are cold-driven rather than drought-related based on environmental records.

3.2.2. Identified gap

The field-only design lacks controlled-environment

manipulations to disentangle temperature/freezing effects from photoperiod, dormancy state, and other seasonal variables. While Fig. S2 and Fig. S4 show robust seasonal separations and Table 1 quantifies many DAPs, field winter differs from summer in multiple variables beyond temperature (photoperiod, developmental state, soil biota). Without chamber validation holding photoperiod and moisture constant while varying temperature/freezing, attributing seasonal differences specifically to cold exposure remains inferential. This is especially relevant given the phrasing that changes occur under freezing conditions.

3.2.3. Suggested textual edits

First, justify the current findings by emphasizing documented cold exposure: add "winter collections followed sustained sub-zero exposure reaching -16°C and soil temperatures below $2-5^{\circ}\text{C}$ for 20-22 days." Next, acknowledge the limitation: "Field winter and summer differ in temperature, photoperiod, dormancy state, and soil biota; we cannot fully isolate temperature effects without chamber validation." Finally, modify the interpretation: Replace "establishing these seasonal contrasts as cold-driven rather than drought-related." with "consistent with cold-associated changes while other seasonal factors may contribute."

3.2.4. Potential next steps

To address the field-only design and causal attribution to cold:

3.2.4.1. Option 1

Perform cross-omics enrichment by testing whether

winter-elevated rhizome proteins (e.g., LEA3, HSPs, GSTs) are enriched among cold-upregulated genes in existing 5°C vs 22°C *Tripsacum* RNA-seq datasets (seedling leaves and 'Pete' crown tissues). This will provide independent evidence that a subset of field responses are temperature-driven.

3.2.4.2. Option 2

Conduct controlled-environment rhizome proteomics in growth chambers using clonal *P. virgatum* and *A. gerardi* in a factorial design separating temperature (22°C vs 5°C) and photoperiod (12 h vs 8 h), with optional freeze events. Analyze TMT-labeled samples on the Orbitrap Eclipse and use two-way ANOVA to quantify cold- versus photoperiod-specific proteome changes.

3.3. Related claim 1.1

LEA3 is the only ortholog universally elevated among the top winter proteins across all five species

3.3.1. GAP 1 (MINOR) Uneven Coverage May Mask Other Universal Orthologs

3.3.1.1. Background information

The assertion that LEA3 is uniquely universal among the top winter-elevated proteins rests on per-species ranked lists derived from rhizome proteomes. These rankings are visualized and summarized in the figures that show functional category mosaics and species-specific top lists (Figure 3a and Figure 3b), with detailed ranks in the supplementary table of top winter proteins (Table S7). The evidence is strong that LEA3 appears in all species' top lists; however, proteome coverage is uneven across species (e.g., fewer identifications in *Sorghastrum nutans*

and proteome remapping for *Miscanthus*), which could influence which orthologs reach the "top" threshold.

3.3.1.2. Identified gap

Uneven proteome coverage and annotation across species may have suppressed detection or ranking of other universally winter-elevated orthologs, biasing the conclusion that LEA3 is uniquely universal.

Figure 3a, Figure 3b, and Table S7 depend on species-specific detection and annotation; Table 1 and Table S2 indicate that *Sorghastrum nutans* had fewer identifications and a draft genome with high ambiguity, and *Miscanthus* required peptide remapping. Proteomics studies consistently show detection bias toward well-annotated, highly expressed proteins, with only 35% validation of annotated genes and systematic under-detection of lowly expressed proteins [1][2]. Comparative proteomics exhibits bias toward conserved isoforms and suffers from 'one-hit-wonder' detection limitations that can miss true orthologs [1][3]. In this context, absence of a protein from a species' top list could reflect under-detection or misannotation rather than true biological absence. Without harmonized coverage (e.g., DIA with species-specific libraries) or targeted validation, the 'only ortholog' uniqueness may partly reflect technical disparities shown alongside the ranking visualizations in Figure 3 and the dataset summaries in Table 1.

3.3.1.3. Suggested strategies

3.3.1.3.1. Suggested textual edits

First, justify the current findings by clarifying

the detection basis and consistency of LEA3 across species. For example, add: "Notably, LEA3 was detected and winter-elevated in all species despite variation in proteome completeness (Table S2), reinforcing its conserved prominence." Next, acknowledge the limitation: "Per-species top lists may be influenced by detection depth and annotation differences; therefore, 'universally among the top winter proteins' pertains to proteins detected across all species, and additional depth may reveal further universally elevated orthologs." Finally, modify the claim to reflect this nuanced perspective: Replace "LEA3 (OG0022470) was the only ortholog ranked among the top 10 winter-elevated proteins in all five species and was the single most elevated protein in *Tripsacum* ($\log_2\text{FC} = 5.0$; Fig. 3b; Table S7)." with "LEA3 (OG0022470) was the only ortholog among the top winter-elevated proteins across all five species within the shared detection set and was the most elevated in *Tripsacum* ($\log_2\text{FC} = 5$)." ."

3.3.1.4. Potential next steps

To address uneven proteome coverage potentially biasing the conclusion that LEA3 is uniquely universal:

3.3.1.4.1. Option 1

Perform detection-matched re-ranking by restricting to the intersection of orthogroups detected in both seasons across all five species. Quantile-normalize per-species $\log_2\text{FC}$ s within this shared set, compute within-species robust z-

score ranks, and aggregate across species using a rank-product statistic. In parallel, conduct rarefaction: downsample each species' detected proteins to the minimum detection count (*S. nutans*) and recompute top lists (top 10, top 5%) over 1,000 bootstrap iterations. Include EF1 orthologs as negative controls and GST/HSP1/PEBP as comparators. This will provide a detection-equalized assessment of whether LEA3 is uniquely present among universal top winter proteins.

3.3.1.4.2. Option 2

Generate depth-equalizing TMTpro 18-plex batches for species with lower IDs (e.g., *S. nutans*, *M. giganteus*), using the established phenol/S-Trap workflow, with increased fractionation (e.g., 20 pools) and optimized Orbitrap Eclipse RTS-SPS-MS3 settings (narrower isolation windows, longer injection times). Reapply DIAMOND-based peptide remapping as used for *Miscanthus* to other species as needed. Recompute top lists and universality after harmonizing depth. This establishes whether any additional orthologs consistently enter the top winter-elevated set when coverage disparities are minimized.

4. Main Claim 2

Shared cold-responsive proteins show evolutionary constraint on response magnitude with higher cross-species fold-change correlation than background proteins

4.1. GAP 1 (MAJOR) Correlation Contrast Unmatched for Effect Size

4.1.1. Background information

The cross-species conservation analysis hinges on

comparing correlations for orthogroups that are significantly seasonal in at least two species against those that are not, as shown in the boxplots of correlations and the species-pair scatterplots (Figure 2b and Figure 2a). The elevated median correlation for the shared set is striking and strongly suggests a patterned response across lineages. This aspect is central to inferring evolutionary constraint on response magnitude from the observed higher correlations.

4.1.2. **Identified gap**

The comparison between shared seasonally regulated orthogroups and background does not control for fold-change magnitude and cross-species detectability, which can inflate correlations for the shared set. Orthogroups meeting significance in multiple species are enriched for large and consistently directional fold-changes, whereas the background includes many near-zero changes; this asymmetry can artificially increase correlation for the shared set (Figure 2b) relative to a mismatched control. Gene set analysis methods have established approaches to address such correlation inflation through variance inflation factors and proper null models [1], and comparative transcriptomics studies routinely implement magnitude-matched controls and cross-species detectability corrections [2]. Without magnitude- and detection-matched nulls, the observed difference in correlations could reflect selection of high-signal proteins rather than evolutionary constraint (Figure 2a-b). Incorporating matched resampling or permutation tests would clarify whether the correlation contrast persists beyond effect-size selection.

4.1.3. **Suggested strategies**

4.1.3.1. **Suggested textual edits**

First, justify the current findings by noting the stringent thresholds and detection filters already applied. For example, add: "Our stringent $|\log_2\text{FC}|$ threshold and requirement of detection in ≥ 4 species reduce spurious conservation due to low signal and missingness." Next, acknowledge the limitation: "Selecting orthogroups significant in multiple species enriches for large effects and detectability, which can inflate correlations relative to background. Magnitude- and detectability-matched nulls would further test robustness." Finally, modify the claim to reflect this nuanced perspective: Replace "This pattern-tight constraint on response dynamics but flexibility in baseline levels-suggests that independent cold adaptations are governed by conserved regulation of a core set of protective proteins." with "This pattern-tight constraint on response dynamics but flexibility in baseline levels-is consistent with conserved regulation of a core set of protective proteins, although effect-size selection may contribute."

4.1.3.2. **Potential next steps**

To address the comparison between shared seasonally regulated orthogroups and background lacking effect-size/detectability control:

4.1.3.2.1. Option 1

Implement matched resampling of background orthogroups to the shared set on $|\log_2\text{FC}|$ bins, cross-species detectability (≥ 4 vs 5-of-5), PSM intensity, and replicate CV; run 1,000-10,000

resamples and permutation tests that shuffle orthogroup labels within species; perform leave-one-species-out prediction. This will provide effect-size/detectability-controlled nulls and an out-of-sample test of cross-species coherence.

4.1.3.2.2. Option 2

Expand to a hierarchical model that estimates an orthogroup-level 'constraint score' as the variance of log2FC across species after controlling for mean |log2FC| and detectability, with uncertainty estimated by bootstrap. Combine with five-of-five detection sensitivity analyses and cross-validated prediction to derive a composite evidence score for constraint. This establishes whether the correlation advantage is robust to multiple controls and modeling choices.

4.2. **GAP 2 (MINOR)** No Cold-Sensitive Controls or Tolerance Calibration

4.2.1. Background information

All species were exposed to the same field winter conditions, and the correlation analyses were performed on season contrasts from this common garden (Figure 1b and Figure 2). The phylogenetic context (Figure 1e) supports independent evolution, strengthening the comparative rationale. However, attributing the observed correlation to cold-adaptive constraint rather than broadly shared seasonal physiology depends on the experimental context.

4.2.2. Identified gap

The study lacks cold-sensitive or tropical relatives as negative proteomic controls and does not calibrate cold exposure to each species' tolerance, limiting

attribution of higher correlations to cold adaptation per se. Comparative proteomic studies routinely include cold-sensitive lineages as controls [1][2] and calibrate exposures to species-specific tolerance thresholds [3][4] to distinguish adaptive responses from general seasonal programs. Without including non-cold-adapted lineages or matching cold intensity to species-specific tolerance (e.g., relative to LT50), similar correlations could arise from shared winter dormancy programs rather than adaptive constraint (Figure 1b, Figure 1e, Figure 2a-b). The authors note the identical-winter-condition limitation, but adding negative lineages or tolerance-calibrated exposures would more firmly connect the correlation pattern to the evolution of cold tolerance rather than general seasonality.

4.2.3. Suggested strategies

4.2.3.1. Suggested textual edits

First, justify the current findings by emphasizing functional signals already observed. For example, add: "Winter-elevated proteins were enriched for folding, stress response, and ROS scavenging, and showed greater conservation than winter-reduced proteins." Next, acknowledge the limitation: "We did not include proteomic data from tropical or cold-sensitive relatives; general winter dormancy may contribute to conserved patterns under identical field winters." Finally, modify the claim to reflect this nuanced perspective: Replace "The most striking finding is the evolutionary constraint on response magnitude." with "The most striking finding is an apparent evolutionary constraint on response magnitude." Also replace "Independent evolution of

freezing tolerance in PACMAD grasses is governed by evolutionary constraints on protein-level response magnitude, reflecting the retention of an ancestral protective capacity." with "Independent evolution of freezing tolerance in PACMAD grasses is consistent with evolutionary constraints on protein-level response magnitude, reflecting retention of an ancestral protective capacity."

4.2.3.2. Potential next steps

To address the lack of cold-sensitive/tropical proteomic controls and non-calibrated exposures:

4.2.3.2.1. Option 1

Use existing proteomes to stratify orthogroups into cold-protective vs growth-associated categories and recompute cross-species correlation contrasts within/between these bins; separately quantify conservation for winter-elevated and winter-reduced sets; link fold-change magnitudes to Mesonet-derived winter severity per sampling. These specificity tests will evaluate whether the observed conservation aligns more strongly with cold-protective biology than with general dormancy.

4.2.3.2.2. Option 2

Add a controlled, tolerance-calibrated proteomics experiment including cold-adapted and cold-sensitive lineages. After 5°C acclimation, expose plants to freezing at 80-90% of each species' LT50 (determined by electrolyte leakage), then quantify proteomic responses with TMTpro LC-MS/MS. Comparing cross-species correlations within and across groups under calibrated stress

will directly test whether conserved response magnitude is a feature of cold-adapted lineages.

4.3. **GAP 3 (MAJOR)** No Phylogenetic Control in Correlation Analysis

4.3.1. Background information

The cross-species analysis in Figure 2a computes pairwise Spearman correlations of orthogroup-level winter/summer log2 fold-changes between species, and Figure 2b compares these correlations for shared differentially abundant proteins versus background to infer 'evolutionary constraint' on response magnitude.

4.3.2. Identified gap

The correlation analysis does not adjust for phylogenetic relatedness or the shared common-garden environment and treats species-pair correlations as independent. With five closely related grasses measured under the same seasonal regime, pairwise correlations are not independent, and shared exposure can inflate agreement in fold-changes. Without phylogenetic comparative modeling (e.g., PGLS or independent contrasts) [1][2] or a permutation null that preserves batch and season structure, the higher correlations seen in Figure 2b could reflect environment and lineage structure rather than conserved regulation. Even common garden experiments require phylogenetic correction when analyzing trait correlations across related species [3], as shared evolutionary history creates statistical non-independence that inflates correlation estimates [4].

4.3.3. Suggested strategies

4.3.3.1. Suggested textual edits

First, justify the current computational findings by highlighting the robustness across species pairs.

For example, add: "Seasonal fold-changes were positively correlated across all species pairs ($r = 0.37-0.69$; Fig. 2a)." Next, acknowledge the computational limitation: "Pairwise correlations were not adjusted for phylogenetic relatedness or the common-garden exposure; agreement in fold-changes may partly reflect shared conditions." Finally, modify the computational interpretation to reflect this nuanced perspective: Replace "This pattern⁴tight constraint on response dynamics but flexibility in baseline levels¹suggests that independent cold adaptations are governed by conserved regulation of a core set of protective proteins." with "This pattern is consistent with conserved regulation of a core protective set; confirming evolutionary constraint will require phylogenetic controls and environment-aware nulls."

4.3.3.2. Potential next steps

To address The correlation analysis does not adjust for phylogenetic relatedness or the shared common-garden environment and treats species-pair correlations as independent:

4.3.3.2.1. Option 1

Implement leave-one-species-out stability checks of the $\Delta\Delta$ gap and tip-label permutation on the pruned ASTRAL-Pro tree (10,000 replicates), preserving per-species fold-change distributions and orthogroup presence. Add a block-permutation of season labels within species/TMT batches to generate null distributions for the observed correlation gap. This will provide a baseline

quantifying the contribution of shared exposure and lineage structure to $\Delta\Delta$.

4.3.3.2.2. Option 2

Fit phylogenetic generalized least squares (PGLS) models of orthogroup-level log₂FC across species using the existing tree to estimate phylogenetic covariance, and compute phylogenetic independent contrasts (PIC). Compare shared-DAP versus background concordance under phylogenetic control. This establishes lineage-aware evidence that the higher fold-change correlation among shared DAPs reflects evolutionary constraint rather than solely shared environment.

4.4. **Related claim 2.1** LEA3 subfamily shows consistent winter elevation across species despite early phylogenetic divergence from other LEA clades

4.4.1. No gaps found

4.5. **Related claim 2.2** Five independent evolutionary transitions to freezing tolerance occurred in the study species

4.5.1. No gaps found

5. **Main Claim 3** LEA3 structural divergence in maize compromises cryoprotective function despite transcriptional induction

5.1. **GAP 1 (MINOR)** No Functional Assay of LEA3

Cryoprotection

5.1.1. Background information

The conclusion that motif divergence impairs LEA3 function is based on hydropathy profiling and AlphaFold3 models showing a more hydrophobic third repeat and altered packing in maize, alongside strong LEA3 transcript induction in both *Tripsacum* and maize (Figure 5a-c; Figs. S8-S10; Fig. S7). This is a compelling

mechanistic hypothesis tightly connected to the observed sequence differences. However, the work does not measure the functional cryoprotective activity of the maize variant relative to cold-tolerant orthologs.

5.1.2. Identified gap

The paper lacks direct functional assays comparing cryoprotective efficacy of maize LEA3 to consensus LEA3 proteins. Standard LDH freeze-thaw protection assays have been successfully used to demonstrate LEA3 cryoprotective function in *Arabidopsis* [1] and rice [2], making such comparative studies feasible. Without enzyme- or membrane-protection assays, or transgenic swap tests directly comparing maize LEA3 to consensus sequences, the link from the variant 11-mer repeat to reduced cryoprotection remains inferential. The structural arguments in Figure 5a-c and Figs. S8-S10, and the expression comparisons in Fig. S7, are suggestive but not sufficient to establish causality. This limits the strength of the conclusion that structural divergence, rather than other factors, compromises LEA3 function in maize.

5.1.3. Suggested strategies

5.1.3.1. Suggested textual edits

First, justify the current findings by explicitly linking the structural hypothesis to established LEA mechanisms. For example, add: "Given that Group 3 LEAs shield enzymes and membranes during dehydration and freezing, our motif-level differences provide a mechanistically grounded hypothesis for functional divergence." Next, acknowledge the limitation: "We did not directly measure cryoprotective activity of LEA3 orthologs; enzyme- or membrane-protection

assays and in planta motif swaps are needed to establish causality." Finally, modify the claim to reflect this nuanced perspective: Replace "Structural divergence of LEA3 in maize suggests that transcriptional induction alone does not ensure freezing tolerance; functional protection likely requires intact motif architecture." with "Structural divergence of LEA3 in maize is consistent with the hypothesis that transcriptional induction alone may be insufficient; functional protection may require intact motif architecture."

5.1.3.2. Potential next steps

To address The paper lacks direct functional assays comparing cryoprotective efficacy of maize LEA3 to consensus LEA3 proteins:

5.1.3.2.1. Option 1

Reanalyze the existing TMT-MS3 datasets at the peptide level in Proteome Discoverer to quantify LEA3 peptides spanning the variant third 11-mer repeat versus consensus-repeat peptides. Use reporter-ion intensities from uniquely mapping peptides, normalize to conserved LEA3 peptides and EF1, and compare across species and seasons. Extend AmphipaSeek analyses to compute hydrophobic moments and charge for each 11-mer and relate these to winter log2FC. This will provide direct in vivo confirmation of the variant repeat's accumulation and a quantitative link between motif biophysics and seasonal dynamics.

5.1.3.2.2. Option 2

Functionally compare ZmLEA3, a motif-reverted

ZmLEA3 (third repeat restored to consensus), TdLEA3, and MgLEA3 via LDH protection during freeze-thaw/drying, liposome leakage assays under freeze-thaw, and client-protein aggregation suppression. Include BSA and an unrelated IDP as negative controls and wheat TdLEA3 as a positive control. In planta, express motif-reverted and wild-type ZmLEA3 to test electrolyte leakage and survival after controlled freezing. This will establish whether repeat composition causally affects cryoprotective efficacy and in vivo freezing tolerance.

5.2. **GAP 2 (MINOR)** AlphaFold3 Models Lack Confidence Metrics

5.2.1. Background information

The study uses AlphaFold3 to predict full-length LEA3 structures (Figure 5b) and tetramer assemblies (Fig. S10) and interprets differences in packing geometry as mechanistically relevant. Predictions were run with default parameters and the paper does not report confidence metrics.

5.2.2. Identified gap

The structural inference from AlphaFold3 lacks reporting of confidence scores (e.g., pLDDT, pTM/ipTM) and cross-predictor benchmarking, making conclusions about oligomeric packing and functional differences brittle. For disordered proteins and oligomer assemblies, default predictions can be uncertain; without confidence metrics and replication across AF-Multimer, Rosetta, ESMFold, or repeat runs, the packing differences in Fig. S10 may be artifacts. The absence of quantitative confidence

undermines the robustness of the mechanistic link drawn from Figure 5b and Fig. S10.

5.2.3. Suggested strategies

5.2.3.1. Suggested textual edits

First, justify the current computational findings by clarifying AF3b4s role: "AlphaFold3 provides hypothesis-generating models that we use to contextualize repeat architecture." Next, acknowledge the computational limitation: "We did not report AF confidence metrics (pLDDT, pTM/ipTM) or replicate variability; multimer packing may vary in intrinsically disordered regions." Finally, modify the computational interpretation to reflect this nuanced perspective: Replace "AlphaFold3 multimer predictions reinforce this distinction: while both Td and ZmLEA3 form tetrameric assemblies, they adopt distinct packing geometries (Fig. S10)" with "AlphaFold3 multimer predictions suggest distinct tetrameric packing; reporting pLDDT, pTM/ipTM and replicate variability will contextualize confidence."

5.2.3.2. Potential next steps

To address the lack of AlphaFold3 confidence metrics and cross-benchmarking:

5.2.3.2.1. Option 1

Re-run AlphaFold3 monomer and multimer predictions for Tripsacum and maize LEA3 with multiple random seeds (e.g., N=10). Extract pLDDT (monomer) and pTM/ipTM (multimer), and quantify replicate variability (TM-score/RMSD). This will provide direct model confidence and robustness checks using the same predictor.

5.2.3.2.2. Option 2

Summarize confidence and variability per repeat and per oligomer interface, and integrate these with hydropathy and amphipathicity profiles in R. Report low-confidence regions expected for IDRs and highlight any consistent packing differences with high pTM/ipTM and low replicate variability. This establishes evidence-based structural uncertainty and strengthens the mechanistic narrative.

5.3. **Related claim 3.1** Maize induces LEA3 transcription to levels comparable to cold-tolerant *Tripsacum* yet remains freezing-sensitive

5.3.1. No gaps found